Quadratic equations

1 Solve the following equations:

a $x(x+6)=0$

b $-3m(m+2)=0$

c $x(7x-12)=0$

d $(x-7)(x-6)=0$

e $(x-7)(x+8)=0$

f $(17x-19)(11x-5)=0$

g $(7x+19)(2x-5)=0$

h $(4x-17)(3x+11)=0$

i $(4x-9)^2=0$

j $(3-3x)^2=0$

k $\frac{m}{6}(m-10)=0$

2 Solve the following equations:

a $x^2-64=0$

b $x^2-81=0$

c $10x^2+40x=0$

d $11x^2=7x$

e $81x^2-64=0$

f $6x^2+18=20$

g $x^2+12x=0$

h $3y-15y^2=0$

3 Solve the following equations:

a $x^2+19x+90=0$

b $x^2+6x-55=0$

c $x^2-17x+72=0$

d $x^2-5x-14=0$

e $x^2-20x+100=0$

f $m^2-27m=-182$

g $m^2=m+20$

h $-x^2+x+20=0$

4 Solve the following equations:

a $4x^2+8x-32=0$

b $5x^2+22x+8=0$

c $-4x^2+25x-36=0$

d $4x^2-17x+15=0$

e $4x^2+17x+15=0$

f $5x^2+14x+8=0$

g $3x^2-7x-20=0$

h $4x^2-7x-15=0$

5 Solve the equation $4x^2-x-10=0$ for x, leaving your answer in radical form.

6 Solve the following equations:

a $\dfrac{2x^2 - 19x}{3} = 20$

b $\left(5x^2 + 13x + 6\right)\left(2x^2 + 13x + 20\right) = 0$

c $\dfrac{m}{6}(m - 10) = 0$

d $(4x - 17)(3x + 11) = 0$

e $\dfrac{x^2 - 5x}{8} = 3$

f $(y + 1)^2 = 4y + 4$

7 Solve the following equations by first rearranging, and then factoring:

a $m^2 - 27m = -182$

b $m^2 = m + 20$

c $m^2 = 14m$

d $5m^2 = 7m$

e $(y + 1)^2 = 4y + 4$

f $\dfrac{2x^2 - 19x}{3} = 20$

8 Find the value of a if:

a The nonzero solution of the equation $x(x + a) = 0$ is $x = 10$.

b The nonzero solution of the equation $x(x + a) = 0$ is $x = -1$.

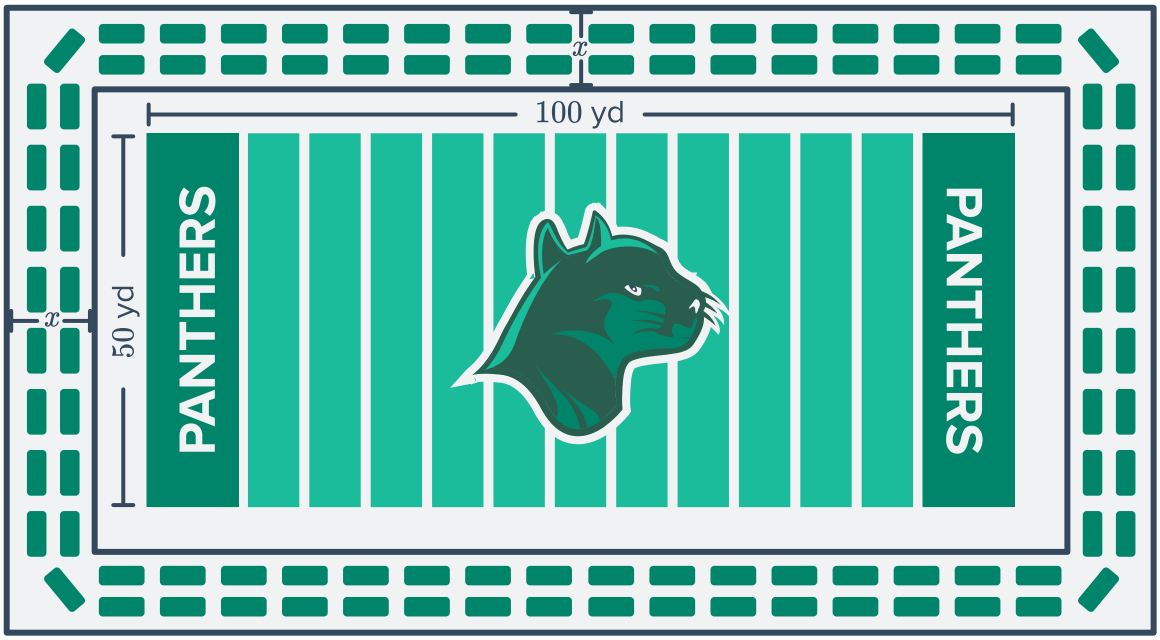
9 A quadratic equation has solutions $x = \pm 2$. What must the equation be?

10 The equation $x(x - 7) = -10$ has a positive integer solution $x = 5$. What is its other solution?

Applications

11 Solve for the x-values of the x-intercepts of the parabola whose equation is $y = x^2 + x - 6$.

12 The school football field is in the shape of a [redacted]ngle and has stadium seating all the way around the field that is of a uniform width.



A = 8400 yd^2

If the total area of the field and the stadium is 8400 yd^2, determine the width of the stadium seating.